

Product Requirement Document (PRD)

Hybrid RAG Engine for E-Commerce Multi-Catalog Support Desk

Document Owner: Lead AI Architect

Target Release: Q3 2026

Status: Approved / Ready for Dev

System Type: Hybrid Retrieval RAG (BM25 + Dense)

Target SLA: < 1.2s End-to-End Latency

Target Accuracy: ≥ 94% Precision@5

1. Executive Summary & Objective

Customer support agents across e-commerce platforms struggle to quickly locate resolution steps for complex technical support tickets. Customer queries naturally mix **fuzzy visual/functional descriptions** (e.g., "squeaking seat recline") with **exact technical identifiers** (e.g., "SKU CHAIR-ERG-X99", "part # P-9913-BASE").

Standard semantic vector search engines fail catastrophically on exact serial number and part identifier matching due to vector embedding smoothing. Conversely, standard keyword search engines (BM25) fail on natural customer phrasing and intent understanding. This project defines the build of an **Enterprise Hybrid RAG System** leveraging Sparse Search (BM25/SPLADE), Dense Embedding Search (OpenAI text-embedding-3-small), Reciprocal Rank Fusion (RRF), and Cross-Encoder Reranking to deliver high-precision support answers.

2. Key Performance Indicators (KPIs)

SKU / PART # EXACT ACCURACY 99.5% Zero hallucination or drop on exact alphanumeric part codes.	SEMANTIC ANSWER RELEVANCE 94.0% High Precision@5 hit rate on complex phrasing queries.
P95 RETRIEVAL LATENCY < 350 ms Combined sparse + dense execution + RRF fusion overhead.	END-TO-END RESPONSE TIME < 1.25 sec Including reranking and streaming LLM token generation.

3. Technical System Architecture

The system executes a multi-stage funnel architecture to guarantee both recall breadth and precision depth.

Hybrid Retrieval Mechanics
Query parsing extracts structured metadata constraints (e.g., Category, Brand) and broadcasts the clean search string to parallel retrieval engines. Candidate outputs from sparse and dense branches are normalized and merged via Reciprocal Rank Fusion (RRF) before passing to a heavy Cross-Encoder.

Pipeline Stage	Technology / Subsystem	Functional Responsibility
1. Pre-Processing	Regex Parser & Query Normalizer	Extracts SKUs, Serial numbers, and part codes into strict filters. Cleans natural text.
2A. Sparse Engine		

Pipeline Stage	Technology / Subsystem	Functional Responsibility
	Qdrant Sparse Vectors / BM25 / SPLADE	Performs inverted-index lexical search for alphanumeric part identifiers and exact tokens.
2B. Dense Engine	Qdrant Dense / text-embedding-3-small	Generates 1536-dim vector embeddings to capture semantic intent and context similarity.
3. Fusion Layer	Reciprocal Rank Fusion (RRF) Engine	Combines candidate sets based on rank positions without requiring score scale alignment.
4. Reranking Layer	Cross-Encoder (ms-marco-MiniLM-L-6-v2)	Performs deep attention-based pairwise scoring on Top 20 candidates down to Top 5.
5. Generation	GPT-4o / Claude 3.5 Sonnet	Constructs grounded, non-hallucinated response with step-by-step resolution & citations.

4. Data Ingestion & Indexing Schema

Product documentation, technical user manuals, and troubleshooting tickets must be indexed with strict JSON structures to enable hybrid retrieval and metadata filtering.

Document Payload Schema (Qdrant Point Structure)

```
{
  "id": "doc_99182_v1",
  "payload": {
    "sku": "CHAIR-ERG-X99",
    "part_numbers": ["P-9912-ARM", "P-9913-BASE", "S-4012-SCRW"],
    "product_name": "ErgoFlex Pro Office Chair",
    "category": "Furniture/Seating",
    "doc_type": "troubleshooting_guide",
    "content": "To fix recline squeaking: locate screw set S-4012-SCRW under base P-9913-BASE. Apply",
    "last_updated": "2026-03-15"
  },
  "vectors": {
    "dense_vector": [0.012, -0.045, ...], // 1536 dimensions
    "sparse_vector": { "indices": [102, 4018, 9081], "values": [1.4, 0.8, 2.1] }
  }
}
```

5. Fusion & Reranking Mathematical Specification

To prevent score mismatch between BM25 (unbounded positive score) and Cosine Similarity ($[-1, 1]$ range), the fusion layer uses **Reciprocal Rank Fusion (RRF)**.

Given a set D of candidate documents and rank positions $r_m(d)$ from retriever m in M :

$$RRF_Score(d) = \sum_{m \in M} \left(\frac{1}{(k + r_m(d))} \right)$$

Where $k = 60$ is a smoothing constant that prevents high-ranking items from dominating excessively. The top 20 candidates sorted by **RRF_Score** are sent to the Cross-Encoder.

6. Detailed Functional Requirements

Req ID	Feature Module	Requirement Specification
FR-01	Hybrid Ingestion Engine	System must automatically process PDF/HTML technical manuals, parse structural tables, extract part SKUs, and build sparse + dense vectors simultaneously.
FR-02	Sub-string SKU Match	System must tokenize hyphenated and alphanumeric strings (e.g. <code>P-9913-BASE</code>) cleanly without discarding special characters.
FR-03	Dynamic Alpha Weighting	Support administrative adjustment of weights between sparse and dense score contributions depending on query intent (e.g., SKU query <code>\$ ightarrow\$ 80% Sparse</code>).
FR-04	Refine & Rerank	Execute cross-encoder reranking on top 20 candidate docs within < 80ms budget.
FR-05	Contextual Citation	LLM output must explicitly tag citations referencing exact manual document IDs and page numbers (e.g. <code>[Doc: CHAIR-ERG-X99, Page 12]</code>).

7. Non-Functional Requirements (NFRs)

- **Scalability:** Vector store must handle 10,000,000 product document chunks across 50,000 SKUs with sub-second retrieval latency.
- **Availability:** Service uptime SLA of 99.95% with blue-green deployment support for index updates.
- **Security & Compliance:** Strict Role-Based Access Control (RBAC); customer ticket content sanitized for PII before embedding generation.
- **Observability:** Integration with OpenTelemetry to track per-stage latency (Sparse Search, Dense Search, RRF, Rerank, LLM time-to-first-token).

8. Implementation & Delivery Roadmap

Phase	Milestone	Key Deliverables
Phase 1	Data Pipeline & Indexing	Build Qdrant hybrid collection, document chunkers, and SKU custom regex tokenizers.
Phase 2	Hybrid Search & RRF	Implement parallel sparse + dense query executor and RRF ranking module.
Phase 3	Reranking & LLM Integration	Add Cross-Encoder model service and integrate streaming output with GPT-4o context guardrails.
Phase 4	Evaluation & Tuning	Run benchmark suite using Ragas / TruLens (Hit Rate@5, MRR, Faithfulness metrics).